

Normalization

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Functional dependency (FD)

Functional Dependency

FD: $x \rightarrow y$

R

x	y
1	1
2	2
3	3
5	5

if $t_1 \cdot x = t_2 \cdot x$
then $t_1 \cdot y = t_2 \cdot y$

Student

R.No	Name	Marks	Dept.	Course
1	a	78	CS	C ₁
2	b	60	EE	C ₁
3	a	78	CS	C ₂
4	b	60	EE	C ₃
5	c	80	IT	C ₃
6	d	80	EC	C ₂

FD: $x \rightarrow y$
 if $t_1.x = t_2.x$
 then $t_1.y = t_2.y$

x y
 $R.No \rightarrow Name$ ✓

$Name \rightarrow R.No$ X

$R.No \rightarrow Marks.$ ✓

$Dept \rightarrow Course$ X

$Course \rightarrow Dept.$ X

$Marks \rightarrow Dept$

$(R.No, Name)$ $\rightarrow Marks$ ✓

x
 $Name \rightarrow Marks.$ ✓

Types of FDs

- Trivial dependency ($AB \twoheadrightarrow A$)
 - $ACD \twoheadrightarrow CD$, $A \twoheadrightarrow A$
- Non trivial dependency ($AB \twoheadrightarrow C$), $XY \twoheadrightarrow Z$
- Multivalued
- transitive

Armstrong rules

1. **Reflexivity:** if B is a subset of A, then $A \rightarrow B$.
2. **Augmentation:** if $A \rightarrow B$ then $AC \rightarrow BC$
3. **Transitivity:** if $A \rightarrow B$ and $B \rightarrow C$ then $A \rightarrow C$.
4. **Self – determination:** $A \rightarrow A$.
5. **Decomposition:** If $A \rightarrow BC$, then $A \rightarrow B$, $A \rightarrow C$.
6. **Union:** if $A \rightarrow B$ and $A \rightarrow C$, then $A \rightarrow BC$
7. **Composition:** if $A \rightarrow B$, $C \rightarrow D$ then $AC \rightarrow BD$.

Attribute closure

- X^+ = set contains all attribute determined by X
- super keys are here = {A,}
- $A^+ = ABCDE$? Sk
- $B^+ = BCDE$? Not sk
- $C^+ = CDE$? not sk
- $D^+ = DE$? not SK
- Replace rule: there is no FD in

which A is present on RHS so stop here.

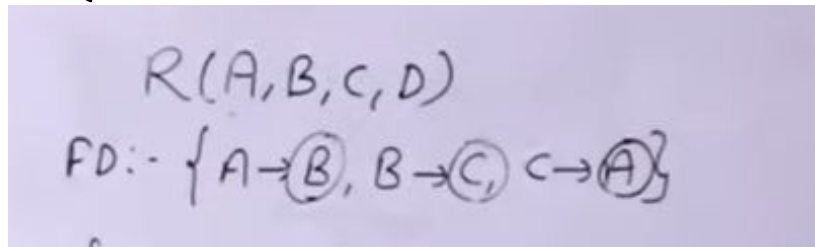
$R(A, B, C, D, E)$
FD: $\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E\}$

$\underline{SK} \leftarrow \underline{A}^+ = \{A, B, C, D, E\}$
 $SK \rightarrow \underline{AD}^+ = \{A, D, B, C, E\}$
 $\times \underline{B}^+ = \{B, C, D, E\}$
 $\times \underline{CD}^+ = \{C, D, E\}$
 $\underline{\underline{AB}}^+ = \{A, B, C, D, E\}$

Super key , candidate key

- If all proper sets of SK are not SK, then that SK is CK
 - If I have AB $\not\subseteq$ sk then I have 2 proper sets of {A, B}

Question :

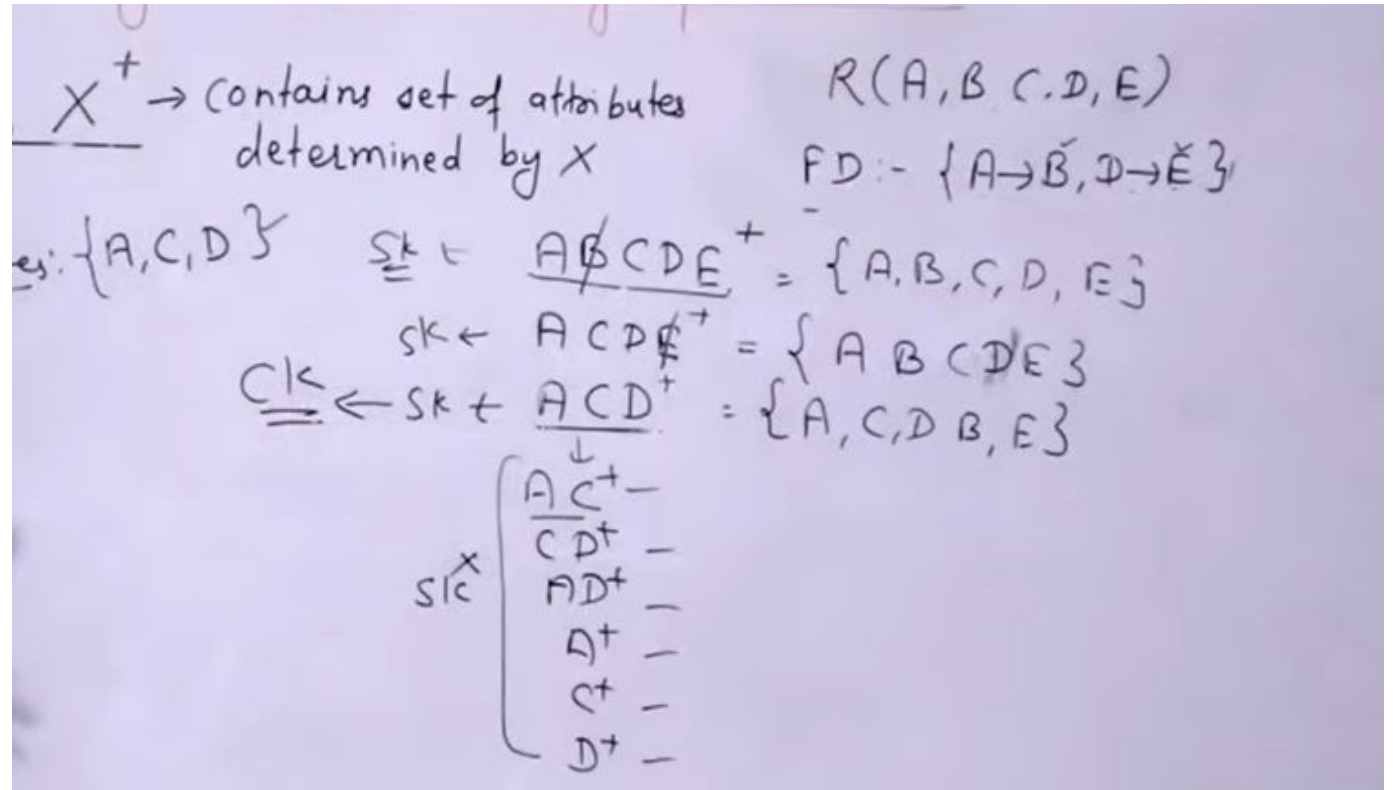


SK={AD,CD,BD}

These are also CK.

Prime attributes = {A,B,C,D}

NPA = {null}



$$AD \not\subseteq \overline{CD}^+ = \overline{BD}^+ = AD$$

Replacement of keys

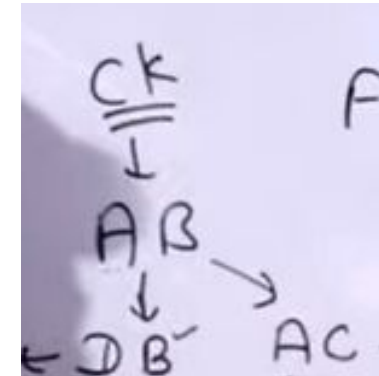
- $Sk = \{AB, C, DB\}$
 - $DB+ = DBACEF \oplus sk$
 - $CK = \{AB, C, DB\}$
 - $Prime = \{A, B, C, D\}$
 - $NPA = \{E, F\}$
-
- $AB+ = ABCDEF \oplus SK$
 - $DB+ = DBACEF \oplus sk$
 - $AC+ = ACDEBF \oplus sk$
 - $C+ = CDEBFA \oplus sk$
 - $CD+ = CDEFBA \oplus SK$

Finding all Candidate keys in a relation

$R(A, B, C, D, E, F)$

$F = \{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, D \rightarrow A, C \rightarrow B\}$

AB, AD
BC, CD

$$\begin{aligned} AB\bar{C}DEF^+ &= \{A, B, C, D, E, F\} \\ AB\bar{D}EF^+ &= \{A, B, C, D, E, F\} \\ AB\bar{F}^+ &= \{A, B, C, D, E\} \\ C\bar{A}B^+ &= \{A, B, C, D, E, F\} \\ \downarrow \\ \{A\}^+ &= \{A\}^* \\ \times \{B\}^+ &= \{B\}^* \\ \{D\}^+ &= \{D\}^* \\ \{C\}^+ &= \{C, D, E, F, A, B\} \end{aligned}$$


Canonical /Minimal cover

- Step1: split RHS of each FD into single attribute
- Step2: remove extra attributes at LHS of each FD
- Step3: remove redundant and trivial FDs
- Step4: take closure of remaining, if attribute come in result ,discard it otherwise keep it.

Minimal cover example

- Step1: split RHS

$$F: - \{ AB \rightarrow C, C \rightarrow AB, B \rightarrow C, ABC \rightarrow AC, A \rightarrow C, AC \rightarrow B \}$$

$$F_1: = \{ AB \rightarrow C, C \rightarrow A, C \rightarrow B, B \rightarrow C, ABC \rightarrow A, ABC \rightarrow C, A \rightarrow C, AC \rightarrow B \}^u$$

$$= \{ \dots \}$$

- Step 2: removed trivial

$$\{ AB \rightarrow C, C \rightarrow A, C \rightarrow B, B \rightarrow C, \cancel{ABC \rightarrow A}, \cancel{ABC \rightarrow C}, A \rightarrow C, AC \rightarrow B \}^u$$

- $AB \twoheadrightarrow C, C \twoheadrightarrow A, C \twoheadrightarrow B, B \twoheadrightarrow C, A \twoheadrightarrow C, AC \twoheadrightarrow B$

- Step3: remove extra attribute on lhs

$$= \{ \cancel{AB} \rightarrow C, C \rightarrow A, C \rightarrow B, B \rightarrow C, \cancel{ABC \rightarrow A}, \cancel{ABC \rightarrow C}, A \rightarrow C, AC \rightarrow B \}^u$$

$$= \{ A \rightarrow C, C \rightarrow A, C \rightarrow B, B \rightarrow C, A \rightarrow C \}$$

Solution :

$AB \twoheadrightarrow C, C \twoheadrightarrow A, C \twoheadrightarrow B, B \twoheadrightarrow C, A \twoheadrightarrow C, AC \twoheadrightarrow B$

$B \twoheadrightarrow C, C \twoheadrightarrow A, C \twoheadrightarrow B, B \twoheadrightarrow C, A \twoheadrightarrow C, AC \twoheadrightarrow B$

$B \twoheadrightarrow C, C \twoheadrightarrow A, C \twoheadrightarrow B, A \twoheadrightarrow C, A \twoheadrightarrow B$

$B^+ = B, C^+ = CB, C^+ = CAB$

$A^+ = ABC, A^+ = A$

$B \twoheadrightarrow C, C \twoheadrightarrow A, A \twoheadrightarrow B$

- Other way:

$\{ AB \twoheadrightarrow C, C \twoheadrightarrow AB, B \twoheadrightarrow C, ABC \twoheadrightarrow AC, A \twoheadrightarrow C, AC \twoheadrightarrow B \}$
 $\{ \cancel{AB \twoheadrightarrow C}, C \twoheadrightarrow A, \underline{C \twoheadrightarrow B}, B \twoheadrightarrow C, \cancel{ABC \twoheadrightarrow A}, \cancel{ABC \twoheadrightarrow C}, A \twoheadrightarrow C, \cancel{AC \twoheadrightarrow B} \}$
 $\{ B \twoheadrightarrow C, C \twoheadrightarrow A, C \twoheadrightarrow B, B \twoheadrightarrow C, A \twoheadrightarrow C, A \twoheadrightarrow B \}$

- Remove redundant FDs

$\{ \cancel{B \twoheadrightarrow C}, C \twoheadrightarrow A, C \twoheadrightarrow B, B \twoheadrightarrow C, A \twoheadrightarrow C, A \twoheadrightarrow B \}$

- Closure : $C \twoheadrightarrow A, B \twoheadrightarrow C, A \twoheadrightarrow B$

Equivalent FD sets

- In DBMS, Two different sets of functional dependencies for a given relation may or may not be equivalent.
- If F and G are the two sets of functional dependencies, then following 3 cases are possible-
 - **Case-01:** F covers G ($F \supseteq G$)
 - **Case-02:** G covers F ($G \supseteq F$)
 - **Case-03:** Both F and G cover each other ($F = G$)

- **Step-01:**

- Take the functional dependencies of set G into consideration.
- For each functional dependency $X \rightarrow Y$, find the closure of X using the functional dependencies of set G.

- **Step-02:**

- Take the functional dependencies of set G into consideration.
- For each functional dependency $X \rightarrow Y$, find the closure of X using the functional dependencies of set F.

- **Step-03:**

- Compare the results of Step-01 and Step-02.
- If the functional dependencies of set F has determined all those attributes that were determined by the functional dependencies of set G, then it means F covers G.
- Thus, we conclude F covers G ($F \supseteq G$) otherwise not.

- **Case-03: Determining Whether Both F and G Cover Each Other-**

- If F covers G and G covers F, then both F and G cover each other.
- Thus, if both the above cases hold true, we conclude both F and G cover each other ($F = G$).

Problem

- relation R (A , C , D , E , H) is having two functional dependencies sets F and G as shown:

$F = \{ A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H \}$

$G = \{ A \rightarrow CD, E \rightarrow AH \}$

Solution: F covers G

- $(A)^+ = \{ A, C, D \}$ // closure of left side of $A \rightarrow CD$ using set G
- $(E)^+ = \{ A, C, D, E, H \}$ // closure of left side of $E \rightarrow AH$ using set G
-

- **Step-02:**

- $(A)^+ = \{ A, C, D \}$ // closure of left side of $A \rightarrow CD$ using set F
- $(E)^+ = \{ A, C, D, E, H \}$ // closure of left side of $E \rightarrow AH$ using set F

G covers F

- **Step-01:**

-

- $(A)^+ = \{ A , C , D \}$ // closure of left side of $A \rightarrow C$ using set F

- $(AC)^+ = \{ A , C , D \}$ // closure of left side of $AC \rightarrow D$ using set F

- $(E)^+ = \{ A , C , D , E , H \}$ // closure of left side of $E \rightarrow AD$ and $E \rightarrow H$ using set F

-

- **Step-02:**

-

- $(A)^+ = \{ A , C , D \}$ // closure of left side of $A \rightarrow C$ using set G

- $(AC)^+ = \{ A , C , D \}$ // closure of left side of $AC \rightarrow D$ using set G

- $(E)^+ = \{ A , C , D , E , H \}$ // closure of left side of $E \rightarrow AD$ and $E \rightarrow H$ using set G

Task 1:

- https://www.tutorialspoint.com/dbms/equivalence_of_functional_dependency.htm
- **Q.2 Let us take another example to show the relationship between two FD sets. A relation $R2(A,B,C,D)$ having two FD sets**
- **$FD1 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C\}$ and $FD2 = \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$**

Normal Forms

- **First Normal Form (1NF):**

- **Eliminating Duplicate Records**

- A table is in [1NF](#) if it satisfies the following conditions:
 - All columns contain atomic values (i.e., indivisible values).
 - Each row is unique (i.e., no duplicate rows).
 - Each column has a unique name.
 - The order in which data is stored does not matter.

Second Normal Form (2NF):

- **Eliminating Partial Dependency**

- A relation is in [2NF](#) if it satisfies the conditions of 1NF and additionally.
- No partial dependency exists, meaning every non-prime attribute (non-key attribute) must depend on the entire primary key, not just a part of it.
- $PA \nrightarrow NPA$
- PD = Proper subset of CK $\rightarrow$ Non-Prime Attribute
 - $AB \rightarrow C$ true ck = { AB }
 - $A \rightarrow C$ (PD) table is not in 2nd NF

Third Normal Form (3NF):

- **Eliminating Transitive Dependency**

- A relation is in 3NF if it satisfies 2NF and additionally, there are no transitive dependencies. In simpler terms, non-prime attributes should not depend on other non-prime attributes.
- $TD = NPA \twoheadrightarrow NPA$ should not hold
 - $E \twoheadrightarrow F$ ck= AB
 - $EA \twoheadrightarrow F$ (TD)
 - $E \twoheadrightarrow AF$ (TD)
 - $E \twoheadrightarrow A, E \twoheadrightarrow F$

Boyce-Codd Normal Form (BCNF):

- **The Strongest Form of 3NF**

- BCNF is a stricter version of 3NF where for every non-trivial functional dependency ($X \rightarrow Y$), X must be a superkey (a unique identifier for a record in the table).

- $Ck = \{AB, C, DB\}$

- Check 2nd NF:

- $AB \twoheadrightarrow C$ (not PD) True
- $C \twoheadrightarrow DE$ (not PD) true
- $E \rightarrow F$ (not PD) true
- $D \twoheadrightarrow A$ (not PD) true
- $C \twoheadrightarrow B$ (not PD) true

Handwritten notes on a light blue background. The title is "Finding all Candidate keys in a relation" written in red cursive. Below it, the relation is defined as $R(A, B, C, D, E, F)$. The functional dependencies are listed as $F = \{AB \rightarrow C, C \rightarrow DE, E \rightarrow F, D \rightarrow A, C \rightarrow B\}$. To the right, a box contains the candidate keys: AB, AD on the top line and BC, CD on the bottom line.

Highest normal

	AB $\rightarrow$ C	C $\rightarrow$ DE	E $\rightarrow$ F		
BCNF	true	false	true		
3 rd NF					
2 nd NF					

R(ABC)

FD: { A $\rightarrow$ B, B $\rightarrow$ C, C $\rightarrow$ A }

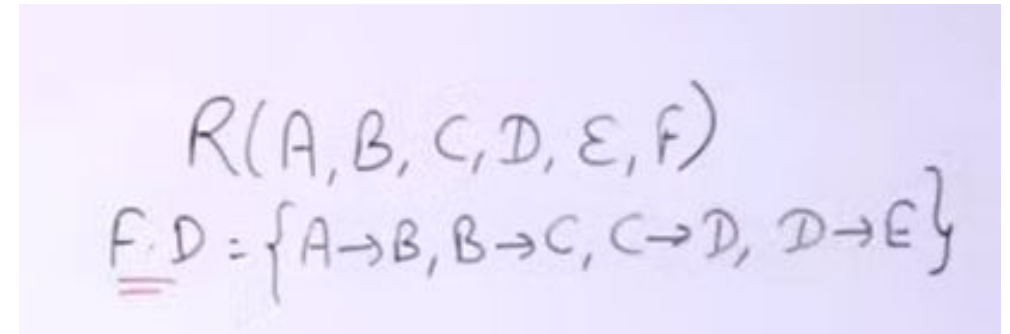
Ck= {A,B,C}

RULES

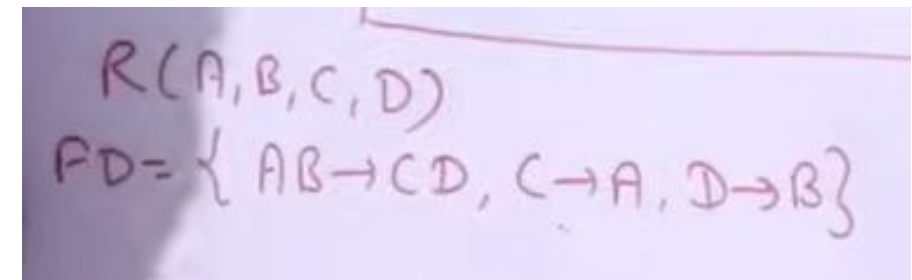
- 2nd NF:
 - PD = {proper subset of CK $\cap$ non prime attribute}
- 3rd NF:
-
- BCNF:

Example of 2nd NF:

- CK = {AF}
- pA={ A,F} NPA={B,C,D,E}
- first fd is PD
 - $A \twoheadrightarrow B$
 - Rest all FDs are in 2nd normal form
 - Overall this relation is not in 2nd normal form
 - Highest normal form is 1st.
- In 2nd question:
- Ck= {AB,BC,AD}
 - This Relation in 2nd NF.



$R(A, B, C, D, E, F)$
F.D = $\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow E\}$



$R(A, B, C, D)$
FD = $\{AB \rightarrow CD, C \rightarrow A, D \rightarrow B\}$

- CK={A}
- Note:
 - If all CKs are single attribute then relation is in 2nd NF
 - Why?

$R(A, B, C, D) \checkmark$
FD: - $\{ \underline{A} \rightarrow B, B \rightarrow C, C \rightarrow D \}$

Example of 3rd NF

Q1: R(ABCD)

- FD={A→B, B→C, C→D}
- CK={A}
- In 2nd NF not 3rd

Q2: R(ABCDEF)

- FD={AB→CDEF, BD→F}
- CK={AB}
- In 2nd fd, D is non prime, therefor lhs is considered as non prime.
- Not in 3rd NF

A table is in 3NF if and only if for each of its non-trivial functional dependency, at least one of the following conditions holds:-

- ① L.H.S. is SK
- ② R.H.S is Prime attribute.

Example of BCNF

- R(ABC)
 - FD={A→B, B→C, C→A}
 - Ck={A,B,C}
- R(ABCDE)
 - FD={A→BCDE, BC→ACE, D→E}
 - Ck={A,BC}
 - Not in BCNF
- Highest NF?

BCNF :-	✓	✓	✗
<u>3NF</u>	✓	✓	✗
<u>2NF</u>	✓	✓	✓

Example (solve)

- R(ABCDE)
 - FD={AB $\rightarrow$ CDE, D $\rightarrow$ A}
 - Ck={?}
 - Highest NF?

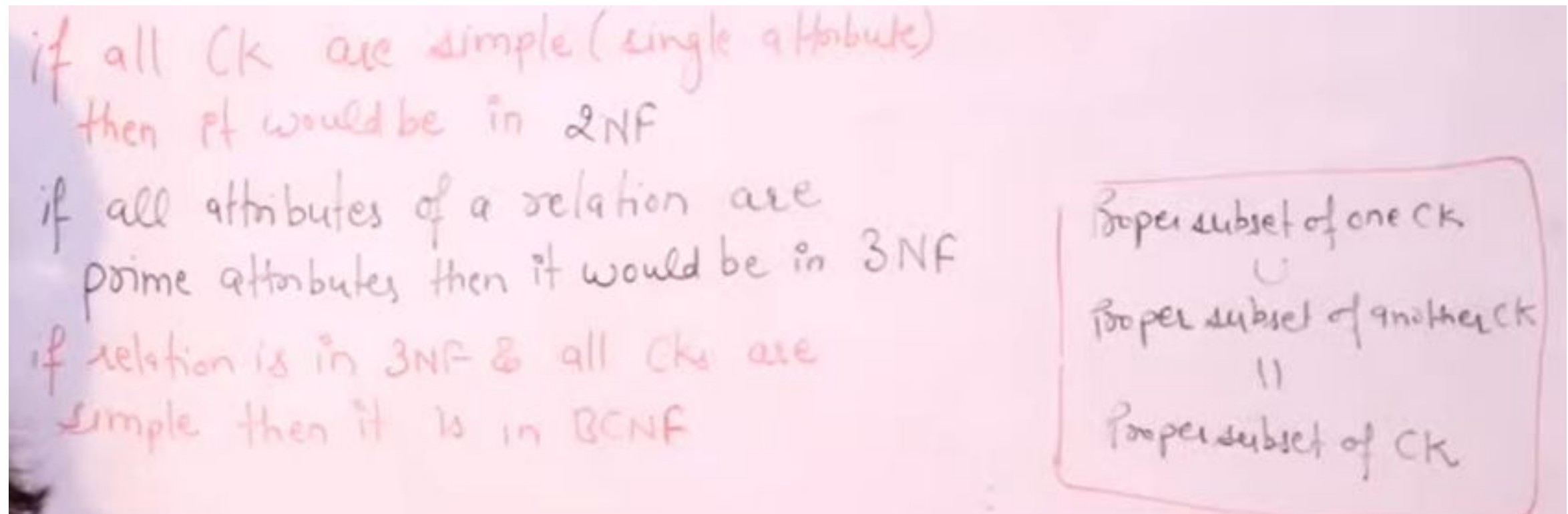
Sample examples

Identify highest Normal Form
 $R(ABCDEFGH)$
 $F:- (ABC \rightarrow DE, E \rightarrow GH, H \rightarrow G, G \rightarrow H, ABCD \rightarrow EF)$

$R(ABCD)$
 $F:- \{AB \rightarrow CD, AC \rightarrow BD, BC \rightarrow D\}$

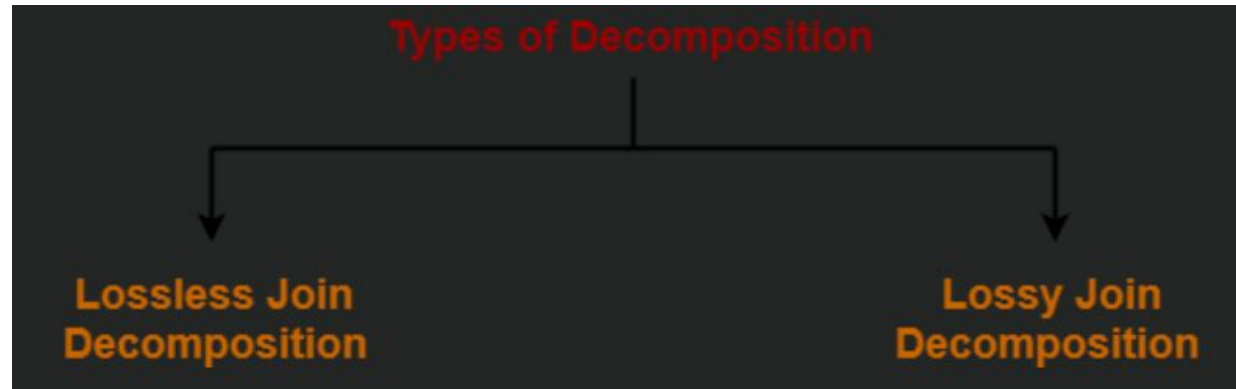
Important tips to remember

- Proven:



Normalization

- Decomposition is a process of dividing a single relation into two or more sub relations.
 - Before splitting table into two or multiple tables one must keep in mind that decomposition should hold 2 properties:



- $R_1 \cup R_2 = R$
- $R_1 \cap R_2 \neq \emptyset$
- $R_1 \cap R_2 = \text{Super key of } R_1 \text{ or } R_2$

Lossless join

- intersection of both the sub relations must be the super key of one of the two sub relations or both.
- $R_1(ABC) \ R_2(CDE)$
 - Common column is C
 - C should be SK/CK of any of r1 or r2

Given decomposition is wrong why?

- n

$R(A, B, C)$	$R_1(A, B)$	$R_2(B, C)$
$\begin{array}{ccc} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \\ 4 & 2 & 2 \end{array}$	$\begin{array}{cc} 1 & 1 \\ 2 & 1 \\ 3 & 2 \\ 4 & 2 \end{array}$	$\begin{array}{cc} 1 & 1 \\ 1 & 2 \\ 2 & 1 \\ 2 & 2 \end{array}$
FD: $\{A \rightarrow BC, BC \rightarrow A\}$	FD(f_1): $\{A \rightarrow B\}$	$f_2: - \{ \}$

Check dependency preserving?

- Compute FDs of sub relations

$R(A, B, C, D, E)$
 $F: - \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$

$R_1(A, B, C)$ $A^+ = ABC$ $B^+ = BCA$ $C^+ = CAB$ $AB^+ = ABC$ $BC^+ = BCA$	$R_2(C, D, E)$ $C^+ = CDE$ $D^+ = DEC$ $E^+ = E$ $CD^+ = CDE$ $DE^+ = DEC$ $CE^+ = CED$
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$F_1 = A \rightarrow BC, B \rightarrow CA, C \rightarrow AB$
 $F_2 = C \rightarrow D, D \rightarrow C$
 $F_1 \cup F_2 =$

$$F = \{A \rightarrow BC, B \rightarrow CA, C \rightarrow AB, C \rightarrow D, D \rightarrow C\}$$

Problems

- Consider a relation schema $R (A , B , C , D)$ with the functional dependencies $FD:\{A \rightarrow B \text{ and } C \rightarrow D\}$. Determine whether the decomposition of R into $R_1 (A , B)$ and $R_2 (C , D)$ is lossless or lossy.
- Consider a relation schema $R (A , B , C , D)$ with the following functional dependency $FD:\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow B\}$. Determine whether the decomposition of R into $R_1 (A , B)$, $R_2 (B , C)$ and $R_3 (B , D)$ is lossless or lossy.

Convert 1NF to 2nd NFx

- R(ABCD)
 - FD: {A → B, B → C} CK={AD} PA={A,D}
 - Fd1 is PD A → B × take closure of that FD
 - A⁺ = ABC

Handwritten notes showing the decomposition of a relation R(ABCD) into two relations R₁ and R₂ to achieve BCNF.

Left side (R₁):

- $A^+ = \underline{ABC}$
- $R_1 = \{A, B, C\}$
- $A^+ = ABC$
- $F_1 = \{A \rightarrow B, B \rightarrow C\}$
- $A^+ = \cancel{A}BC \checkmark$
- $B^+ = \cancel{B}C$
- $C^+ = \epsilon$
- AD
- $\overline{BC}^+ = \cancel{B}C$

Right side (R₂):

- $R_2 = \{A, D\} \Rightarrow \text{BCNF}$
- $R_2 = \{B, D\}$
- $R_2 = \{C, D\}$
- $F_2 = \{ \}$
- $A^+ = \cancel{A} \cancel{D} \cancel{C}$
- $D^+ = \cancel{D}$

Question:

- Decompose into sub relations

$$\begin{aligned}
 &R(A, B, C, D) \\
 &F = \{ \underline{A} \rightarrow B, C \overset{PD}{\rightarrow} D \} \\
 &A \cancel{B} C D^+ = A B C D \\
 &SK \rightarrow \underline{A C}^+ = \underline{A C B D} \\
 &\quad \hookrightarrow A^+ = \times A B \\
 &\quad \hookrightarrow C^+ = \times C D \\
 &\boxed{C|_K = A C} \\
 &PA = A, C
 \end{aligned}$$

$$\begin{aligned}
 &A^+ = A B \quad C^+ = C D \\
 &R_1 = \{ \underline{A}, B \} \quad R_2 = \{ C, D \} \quad R_3 = \{ A, C \} \\
 &A^+ = A B \quad C^+ = C D
 \end{aligned}$$

$$\begin{aligned}
 &F_1 = \{ A \rightarrow B \} \quad F_2 = \{ C \rightarrow D \} \quad F_3 = \{ \} \\
 &A^+ = A B \quad B^+ = \emptyset \quad A^+ = A B \quad C^+ = C D \\
 &\boxed{F_1 \cup F_2 \cup F_3 \equiv F}
 \end{aligned}$$

2nd NF to 3rd NF

• 1st step:

$$R(A, B, C, D)$$

$$F = \{A \xrightarrow{TD} B, B \xrightarrow{TD} C, C \xrightarrow{TD} D\}$$

$$A \cancel{B} \cancel{C} \cancel{D}^+ = ABCD$$

$$\rightarrow \underline{A}^+ = \underline{ABCD}$$

$$\boxed{CK = A}$$

$$PA = A$$

Final answer

$$\boxed{R_{12}(B, C), R_2(C, D), R_3(A, D)}$$

2nd step:

$$B^+ = \underline{BCD}$$

$$R_1 = \{B, \underline{C}, D\}$$

$$F_1 = \{B \rightarrow CD, C \rightarrow D\}$$

$$C^+ = CD$$

$$R_2 = \{C, \underline{D}\}$$

$$F_2 = \{C \rightarrow D\}$$

$$C^+ = \cancel{C}D$$

$$D^+ = \cancel{D}$$

$$\boxed{B(NF)}$$

$$R_3 = \{A, \underline{B}\}$$

$$F_3 = \{A \rightarrow B\}$$

$$A^+ = \cancel{A}B \cancel{C} \cancel{D}$$

$$B^+ = \cancel{B} \cancel{C} \cancel{D}$$

$$\boxed{R(NF)}$$

3rd step:

$$R_1 = \{B, \underline{C}, D\}$$

$$F_1 = \{B \rightarrow \underline{CD}, C \rightarrow D\}$$

$$B \cancel{C} \cancel{D}^+ = BCD$$

$$B^+ = BCD$$

$$\boxed{CK = B}$$

$$PA = B$$

4th step:

$\underline{CD}$ is still TD. So again decompose

$$C^+ = CD$$

$$R_{11} = \{C, \underline{D}\}$$

$$\boxed{CK = C}$$

$$F_{11} = \{C \rightarrow \underline{D}\}$$

$$C^+ = \cancel{C}D$$

$$R_{12} = \{B, \underline{C}\}$$

$$B^+ = B \cancel{C} \cancel{D}$$

$$C^+ = \cancel{C} \cancel{D}$$

$$F_{12} = \{B \rightarrow C\}$$

$$\boxed{2NF}$$

1st to BCNF

- 1st step:

$$\begin{aligned} R(A, B, C, D, E, F, G, H) \\ F:- \{A \rightarrow BD, B \rightarrow C, E \rightarrow FG, AE \rightarrow H\} \\ ABCDEFGH^+ = ABCDEFGH \\ \boxed{CK = AE} \\ PA = A, E \\ AE^+ = AEBDCFGH \\ \quad \hookrightarrow A^+ = ABDC \\ \quad \hookrightarrow E^+ = EFG \end{aligned}$$

- 2nd step:

$$\begin{array}{c|c|c} A^+ = ABDC & E^+ = EFG & R_3 = \{H, A, E\} \\ R_1 = \{A, B, C, D\} & R_2 = \{E, F, G\} & \end{array}$$

- 3rd step:

2NF $A^+ = ABCD$ $R_1 = \{A, B, C, D\}$ $F_1 = \{A \rightarrow BCD, B \rightarrow C\}$ $CK = A$	$\Sigma^+ = \Sigma FG$ $R_2 = \{\Sigma, F, G\}$ $F_2 = \{\Sigma \rightarrow FG\}$ $CK = \Sigma$ BCNF	$R_3 = \{H, A, E\}$ $F_3 = \{A \rightarrow E\}$ (BCNF) $A \Sigma^+ = A \Sigma B D \phi \phi \phi \phi$ $CK = A \Sigma$
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- 4th step: $B \leq C$ is PD**

$R_1(A, B, C, D)$
 $F_1 = \{A \rightarrow BCD, B \rightarrow C\}$
 $B^+ = B, C$
 $R_{11} = \{B, C\}$
 BCNF

$R_{12} = \{A, D, B\}$
 $F_{12} = \{A \rightarrow BD\}$
 $\boxed{K = A}$
 BCNF